

Experimental Section Restructuring Draft for Tail-Aware Post-Training Quantization

Internal Draft—Not for Submission

I. EXPERIMENTS

A. Experimental and Evaluation Setting

a) Base Model.: We evaluate several pretrained 3D geometry foundation models, including VGGT-1B, Pi3, DUST3R, MAST3R, and OpenD4RT. All models are quantized directly from their pretrained full-precision checkpoints without re-training. VGGT-1B serves as the primary model for the unified quantitative comparison, and its results are reported in the main-result section. The remaining models are used to assess the transferability of TAPTQ across architectures.

b) Tasks and Datasets.: We focus primarily on point-cloud reconstruction. For calibration, we randomly draw 20 samples from the DTU training split and select 8 of them using our progressive calibration-set construction strategy. The resulting calibration set is fixed and shared across all experiments. We evaluate point-cloud reconstruction on 7Scenes and ETH3D, and further assess cross-task robustness through camera-parameter prediction on Co3Dv2. DTU is used as the source of calibration samples rather than as a replacement for the ETH3D test benchmark.

c) Quantization Settings.: We evaluate three representative low-bit configurations, namely W4A8, W6A6, and W8A8, where WbAa denotes *b*-bit weight quantization and *a*-bit activation quantization. W4A8 and W6A6 are the primary settings in the manuscript; W8A8 is retained as an auxiliary validation setting until its protocol-aligned ETH3D result is finalized.

d) Baseline Methods.: We compare TAPTQ with representative post-training quantization methods, including round-to-nearest (RTN), PTQ4ViT, RepQ-ViT, ERQ, GPTQ, SmoothQuant, and QuantVGGT. Unless otherwise stated, the baselines reimplemented within our evaluation framework use the same calibration data, inference settings, and geometry-evaluation protocol as TAPTQ. For each method, we retain its original quantization objective and apply the architecture-specific adaptations required for VGGT. Detailed implementation settings, including quantizer configurations and adaptation strategies, are provided in Appendix A. Results obtained under different evaluation protocols are reported separately and are not used for direct quantitative comparison.

e) Metrics.: For point-cloud reconstruction, we report Acc. and Comp. (lower is better) and N.C. (higher is better). For camera-parameter prediction on Co3Dv2, we report AUC at multiple thresholds (A@30/A@15/A@5/A@3; higher is better).

The table format follows the original manuscript: each row identifies the model, bit-width, and method, followed by mean/median Acc., Comp., and N.C. Lower Acc./Comp.

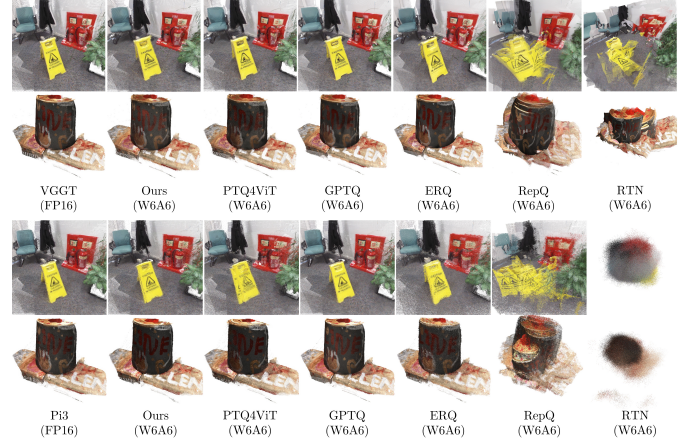


Fig. 1. Qualitative reconstruction comparisons under W6A6. The top panel shows VGGT and the bottom panel shows Pi3; each panel compares the full-precision reference with TAPTQ and representative PTQ baselines.

indicate better reconstruction quality, whereas higher N.C. indicates better normal consistency.

B. Main Results

a) Quantitative Comparison.: We evaluate TAPTQ on point-cloud reconstruction under the primary W4A8 and W6A6 settings. Table I follows the Pi3 evaluation protocol on 7Scenes and the repaired ETH3D 13-sequence $k_f = 5$ protocol. Lower Acc. and Comp. indicate better reconstruction quality, while higher N.C. reflects improved normal consistency.

FP is an independent reference group. Within each W4A8, W6A6, and W8A8 group, boldface marks the best quantized result in each metric column; FP is excluded from this selection. For 7Scenes and ETH3D, QuantVGGT is evaluated with the same full-resolution input construction, frozen frame map, Sim(3)+ICP point-cloud alignment, and Acc./Comp./N.C. computation as the unified evaluator.

b) ETH3D protocol.: ETH3D uses the repaired Pi3-compatible 13-sequence dataset with frozen $k_f = 5$ frame indices and complete RGB/depth/camera triplets. The FP mean Acc. is 0.2909, close to the expected 0.2933 reference scale. Each ETH3D TAPTQ entry uses the selected setting for its bit-width; the corresponding configuration is listed in Table VIII. The W6A6/W8A8 configurations use freshly calibrated checkpoints, not historical records.

The combined qualitative figure complements the aggregate metrics by showing the preservation of surface continuity

TABLE I

VGGT RECONSTRUCTION ON 7SCENES DENSE UNDER THE Pi3 EVALUATION PROTOCOL ($k_f = 40$) AND ETH3D ($k_f = 5, 13$ SEQUENCES). W4A8/W6A6 ARE THE PRIMARY SETTINGS; W8A8 IS AUXILIARY. EACH CELL REPORTS MEAN/MEDIAN. LOWER ACC./COMP. AND HIGHER N.C. ARE BETTER. ETH3D TAPTQ ENTRIES USE THE DATASET-SPECIFIC SELECTED CONFIGURATION; THE ASSOCIATED HYPERPARAMETERS ARE LISTED IN TABLE VIII.

Bit	Method	7Scenes dense ($k_f = 40$)						ETH3D ($k_f = 5$)					
		Acc.↓		Comp.↓		N.C.↑		Acc.↓		Comp.↓		N.C.↑	
		Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.
W4A8	FP	0.0185	0.0156	0.0318	0.0272	0.6895	0.6859	0.2909	0.1985	0.3628	0.2266	0.8433	0.9359
	RTN	0.0317	0.0124	0.0433	0.0158	0.6849	0.7868	0.5522	0.4284	0.6585	0.4059	0.7456	0.8581
	PTQ4ViT	0.0343	0.0152	0.0547	0.0190	0.6877	0.7859	0.5813	0.4490	0.7346	0.4119	0.7290	0.8403
	RepQ-ViT	0.0872	0.0486	0.1939	0.0892	0.6375	0.7073	0.8464	0.7299	2.0768	1.5578	0.5511	0.5829
	ERQ	0.0340	0.0158	0.0367	0.0147	0.6839	0.7857	0.5867	0.4540	0.7044	0.3986	0.7388	0.8470
	GPTQ	0.0304	0.0127	0.0345	0.0152	0.6868	0.7897	0.4322	0.3052	0.5030	0.3082	0.7779	0.8947
	SmoothQuant	0.0337	0.0131	0.0398	0.0139	0.6883	0.7914	0.6618	0.5428	0.8113	0.5057	0.7167	0.8182
	QuantVGGT	0.0207	0.0082	0.0342	0.0176	0.6739	0.7595	0.2765	0.1978	0.3557	0.2339	0.8405	0.9364
	TAPTQ	0.0189	0.0081	0.0359	0.0140	0.6943	0.7983	0.2816	0.1929	0.3457	0.1921	0.8371	0.9367
W6A6	FP	0.0185	0.0156	0.0318	0.0272	0.6895	0.6859	0.2909	0.1985	0.3628	0.2266	0.8433	0.9359
	RTN	0.0420	0.0179	0.0403	0.0157	0.6775	0.7754	0.5261	0.4035	0.6172	0.3913	0.7581	0.8597
	PTQ4ViT	0.0205	0.0092	0.0355	0.0143	0.6918	0.7955	0.4036	0.2869	0.4991	0.3136	0.7886	0.8951
	RepQ-ViT	0.0429	0.0186	0.0455	0.0157	0.6799	0.7788	0.5841	0.4404	0.6938	0.3752	0.7406	0.8457
	ERQ	0.0364	0.0173	0.0407	0.0158	0.6825	0.7830	0.5614	0.4335	0.6615	0.3596	0.7558	0.8631
	GPTQ	0.0404	0.0172	0.0416	0.0150	0.6774	0.7752	0.4758	0.3594	0.6833	0.4487	0.7518	0.8518
	SmoothQuant	0.0328	0.0163	0.0455	0.0178	0.6806	0.7813	0.6009	0.4515	0.7800	0.4911	0.7280	0.8258
	QuantVGGT	0.0203	0.0082	0.0339	0.0171	0.6749	0.7650	0.2929	0.2035	0.3674	0.2321	0.8434	0.9363
	TAPTQ	0.0203	0.0086	0.0349	0.0146	0.6904	0.7935	0.2811	0.1814	0.3287	0.1955	0.8434	0.9441
W8A8	FP	0.0185	0.0156	0.0318	0.0272	0.6895	0.6859	0.2909	0.1985	0.3628	0.2266	0.8433	0.9359
	RTN	0.0216	0.0089	0.0353	0.0166	0.6878	0.7899	0.3572	0.2593	0.5105	0.3423	0.7844	0.8819
	PTQ4ViT	0.0181	0.0074	0.0310	0.0152	0.6899	0.7929	0.2759	0.1779	0.3111	0.1863	0.8456	0.9470
	RepQ-ViT	0.0413	0.0180	0.0410	0.0161	0.6817	0.7810	0.6149	0.4299	0.5957	0.3401	0.7637	0.8844
	ERQ	0.0335	0.0162	0.0363	0.0156	0.7183	0.8077	0.5705	0.4403	0.6088	0.3654	0.7564	0.8681
	GPTQ	0.0213	0.0088	0.0355	0.0167	0.6880	0.7905	0.3718	0.2787	0.5056	0.3292	0.7836	0.8808
	SmoothQuant	0.0230	0.0091	0.0367	0.0170	0.6859	0.7867	0.3525	0.2628	0.4647	0.3018	0.7997	0.8879
	QuantVGGT	0.0205	0.0082	0.0338	0.0173	0.6745	0.7619	0.2908	0.1990	0.3650	0.2275	0.8422	0.9345
	TAPTQ	0.0182	0.0074	0.0311	0.0152	0.6897	0.7926	0.2743	0.1691	0.3146	0.1795	0.8455	0.9468

and geometric structures under low-bit inference across both VGGT and Pi3.

C. C. Cross-Model Transfer

We next evaluate whether the proposed calibration-and-compensation recipe transfers beyond VGGT. Table II reports FP and TAPTQ for each additional model. The rows use each model’s validated preprocessing and module scope, so the table is a transferability check within model families rather than a strict ranking across architectures.

The cross-model results support transferability, but also show that the Acc./Comp./N.C. trade-off is architecture-dependent. We therefore avoid presenting a single setting as universally optimal across all model families.

D. D. Ablation Study

a) *Effect of Calibration Set Construction.*: Table III compares calibration-set construction strategies on 7Scenes under W4A8, using the same initial pool of 20 DTU training samples. “All samples” uses all 20 samples as a high-cost reference, while “rand” reports representative random draws. “Stability score” filters unstable samples before proportional k -means selection. The proposed construction is evaluated at budgets of 4 and 8 samples.

b) *Effect of the Compensation Threshold τ .*: We investigate the threshold used by the TRE-guided compensation module. The W4A8 one-factor sensitivity analysis uses $\rho =$

0.1, $\tau = 0.007$, and $r = 128$, consistently with the W4A8 main-comparison setting. Increasing τ reduces the number of compensated modules, creating a fidelity–overhead trade-off rather than a universal optimum for every metric.

c) *Effect of Ternary Search and Compensation.*: Table IV evaluates ternary search and module-wise compensation under W6A6. Ternary search reduces calibration time while preserving comparable reconstruction quality, and module-wise compensation provides an additional improvement.

d) *Additional Module-Selection and Rank Analysis.*: For completeness, the later artifact-backed analysis compares fixed-budget module selection and the effects of ρ , τ , and rank. These results are reported separately from the main-comparison settings and use the explicitly stated fixed values for each one-factor sweep.

The matched-budget study reports all five completed budgets. TRE is strongest at the reported 48/144 and default 62/144 budgets, while the single-metric optimum can vary at larger budgets; the appropriate conclusion is therefore a favorable accuracy–geometry trade-off rather than universal dominance. The three one-factor hyperparameter sweeps are reported separately in Appendix A-B. Each table states the fixed values used for the other two hyperparameters; the results should therefore be read as independent sensitivity analyses rather than as a full factorial design. ETH3D ablations will be regenerated only after the Pi3-compatible dense-depth evaluation pipeline has been validated.

TABLE II
CROSS-MODEL TRANSFER UNDER W4A8 ON 7SCENES DENSE. RESULTS ARE MEAN/MEDIAN ACC., COMP., AND N.C. UNDER THE E1 FULL18 EVALUATOR. ALL REPORTED ROWS INCLUDE BOTH MEAN AND MEDIAN VALUES.

Model	Method	Acc.↓		Comp.↓		N.C.↑	
		Mean	Med.	Mean	Med.	Mean	Med.
Pi3	FP	0.015044	0.013594	0.022652	0.020248	0.685400	0.675094
Pi3	TAPTQ	0.015167	0.013515	0.019217	0.017985	0.647290	0.642528
DUSi3R	FP	0.019343	0.017784	0.029319	0.023575	0.680308	0.678517
DUSi3R	TAPTQ	0.023192	0.020138	0.027874	0.022460	0.674361	0.667755
MASi3R	FP	0.025488	0.018195	0.031090	0.024432	0.665829	0.664456
MASi3R	TAPTQ	0.032244	0.025711	0.028564	0.022799	0.654313	0.651454
OpenD4RT	FP	0.039151	0.015880	0.049335	0.018543	0.647854	0.729544
OpenD4RT	TAPTQ	0.039429	0.016178	0.048970	0.018367	0.645904	0.726407

TABLE III
ABLATION ON CALIBRATION-SET CONSTRUCTION. ALL METHODS SELECT SAMPLES FROM THE SAME INITIAL POOL OF 20 DTU TRAINING SAMPLES. VALUES ARE MEAN/MEDIAN ACC., COMP., AND N.C. ON 7SCENES DENSE.

Size	Method	Acc.↓		Comp.↓		N.C.↑	
		Mean	Med.	Mean	Med.	Mean	Med.
20	All samples	0.0236	0.0094	0.0361	0.0143	0.6899	0.7930
4	rand (case1)	0.0370	0.0146	0.0580	0.0241	0.6842	0.7835
	rand (case2)	0.0354	0.0158	0.0495	0.0198	0.6864	0.7863
	rand (case3)	0.0332	0.0134	0.0508	0.0178	0.6851	0.7856
	rand (case4)	0.0307	0.0122	0.0468	0.0165	0.6836	0.7819
	Stability score	0.0291	0.0122	0.0470	0.0161	0.6851	0.7852
	TAPTQ	0.0295	0.0121	0.0460	0.0166	0.6868	0.7875
8	rand (case1)	0.0300	0.0131	0.0496	0.0177	0.6868	0.7884
	rand (case2)	0.0342	0.0139	0.0449	0.0158	0.6818	0.7806
	rand (case3)	0.0318	0.0134	0.0478	0.0169	0.6842	0.7841
	Stability score	0.0330	0.0137	0.0436	0.0164	0.6811	0.7805
	TAPTQ	0.0189	0.0081	0.0359	0.0140	0.6943	0.7983

TABLE IV
EFFECT OF TERNARY SEARCH AND MODULE-WISE COMPENSATION UNDER W6A6 ON 7SCENES DENSE. VALUES ARE MEANS; TIME DENOTES CALIBRATION TIME.

Ternary search	Compensation	Acc.↓	Comp.↓	N.C.↑ / Time
Yes	Yes	0.021	0.035	0.692 / 168.7 min
Yes	No	0.023	0.040	0.688 / 161.5 min
No	Yes	0.020	0.036	0.691 / 529.7 min
No	No	0.023	0.039	0.690 / 522.6 min

E. E. Deployment

We report end-to-end deployment measurements for the VGGT-1B aggregator on an NVIDIA B200. Each run processes eight 518^2 frames, uses five warm-up iterations followed by 25 timed iterations, and measures latency with CUDA events. The W8A8 path uses the same `int8_weight_only` backend (`torchao` and `torch.compile(dynamic=True)`) as the low-rank compensation branch. These measurements are reported separately from the algorithmic reconstruction tables.

The low-rank compensation branch is deployed in the same `int8_weight_only` backend. Its per-module overhead is 0.041 ms, corresponding to approximately 2.6% of total latency for the typical 51 compensated modules and 7.5% in the worst-case forced 144-module configuration. Thus, the

compensation module is not merely a calibration-time artifact; its inference overhead remains bounded in the deployed implementation.

APPENDIX A IMPLEMENTATION DETAILS

The appendix discloses the architecture-specific settings used in the experiments. For VGGT, all methods use the DTU-8 calibration set with a 4096-token budget. The ETH3D main-table TAPTQ rows use the selected bit-width-specific configuration listed in Table VIII. The complete mapping from configuration to calibration protocol and result artifact is recorded in the experiment manifest.

A. Supplementary DTU and Configuration Results

Table VII archives the DTU validation results formerly shown in the main table. They are retained as supplementary evidence because the repaired ETH3D $k_f = 5$ protocol is used for the main cross-method comparison.

B. Hyperparameter Ablations

We report the three hyperparameter sweeps as separate one-factor analyses. Each table changes only its first-column variable and fixes the other two VGGT TAPTQ settings in the caption. All entries are evaluated on 7Scenes dense under W4A8; lower Acc./Comp. and higher N.C. are better. The bold parameter value denotes the default value used by the verified ablation suite, while bold metrics denote the best value in that column.

The tail-ratio and rank sweeps favor larger compensation capacity in this artifact-backed setting, while the threshold sweep exposes the expected fidelity–coverage trade-off: $\tau = 0.005$ improves Acc./Comp., whereas the default $\tau = 0.007$ gives the highest N.C. These sensitivity results are not used to replace the main-comparison settings without a protocol-level revalidation.

TABLE V

MATCHED-BUDGET MODULE SELECTION ON VGGT W4A8. RESULTS ON 7SCENES DENSE (MEAN; LOWER ACC./COMP. AND HIGHER N.C. ARE BETTER). ALL ROWS USE THE SAME QUANT-ONLY CHECKPOINT, DTU-8 CALIBRATION, $\rho = 0.1$, AND $\tau = 128$. THE DEFAULT $\tau = 0.007$ SELECTS 62/144 MODULES; THE OTHER BUDGETS USE FIXED-BUDGET TOP- K SELECTION. RANDOM USES SEED 42. BOLD INDICATES THE BEST VALUE WITHIN EACH BUDGET AND METRIC.

Compensated modules	Selection criterion	Acc.↓	Comp.↓	N.C.↑
24/144 (16.7%)	TRE (ours)	0.025874	0.044597	0.688276
	MSE	0.030371	0.045536	0.687632
	Hessian	0.030503	0.046976	0.685933
	Random	0.030941	0.046923	0.688319
48/144 (33.3%)	TRE (ours)	0.021498	0.038211	0.692796
	MSE	0.022470	0.038236	0.691454
	Hessian	0.024763	0.040115	0.689811
	Random	0.022949	0.039355	0.689175
62/144 (43.1%)	TRE (ours; $\tau = 0.007$)	0.019822	0.036430	0.695265
	MSE	0.022227	0.037820	0.690148
	Hessian	0.020257	0.035947	0.692438
	Random	0.022525	0.040430	0.691274
72/144 (50.0%)	TRE (ours)	0.019895	0.035862	0.694169
	MSE	0.020140	0.037739	0.692988
	Hessian	0.019011	0.036116	0.694760
	Random	0.021396	0.038489	0.691965
96/144 (66.7%)	TRE (ours)	0.019795	0.034718	0.694900
	MSE	0.019551	0.036839	0.692821
	Hessian	0.020036	0.036529	0.692752
	Random	0.024370	0.042894	0.689881

TABLE VI

END-TO-END VGGT-1B DEPLOYMENT ON NVIDIA B200 WITH EIGHT 518² FRAMES. LATENCY IS MEASURED AFTER FIVE WARM-UP ITERATIONS OVER 25 TIMED ITERATIONS USING CUDA EVENTS. W DENOTES QUANTIZED WEIGHT STORAGE AND PEAK DENOTES PEAK GPU MEMORY.

Configuration	Latency (ms)	Relative to FP	W (GB)	Peak (GB)
FP32 baseline	401	1.00×	5.0	9.9
W8A8 static + compile	74	5.42×	1.7	4.1
W8-Abf16 + compile	81	4.95×	1.4	3.8

TABLE VII

SUPPLEMENTARY DTU VALIDATION RESULTS FOR VGGT. EACH ENTRY REPORTS MEAN/MEDIAN. LOWER ACC./COMP. AND HIGHER N.C. ARE BETTER.

Bit	Method	Acc.↓		Comp.↓		N.C.↑	
		Mean	Med.	Mean	Med.	Mean	Med.
W4A8	FP	1.1855	0.7145	2.2195	1.3003	0.6893	0.7733
	RTN	1.1822	0.6338	2.2015	0.7478	0.6843	0.7762
	PTQ4ViT	1.1089	0.5998	1.9479	0.7466	0.6836	0.7741
	RepQ-ViT	6.9974	4.8927	11.2423	3.6770	0.6555	0.7268
	ERQ	1.0439	0.5682	1.8350	0.7284	0.6841	0.7735
	GPTQ	1.0436	0.6059	1.8580	0.8658	0.6893	0.7793
	SmoothQuant	1.2738	0.6459	2.1866	0.7530	0.6849	0.7778
	QuantVGGT	1.3056	1.1314	1.4199	0.8459	0.5289	0.4952
	TAPTQ	1.1573	0.6669	2.2094	1.0938	0.6826	0.7686
W6A6	FP	1.1855	0.7145	2.2195	1.3003	0.6893	0.7733
	RTN	1.0875	0.5961	1.9859	0.7216	0.6836	0.7754
	PTQ4ViT	1.2467	0.7020	2.2082	0.9631	0.6780	0.7659
	RepQ-ViT	1.0501	0.5669	1.7998	0.7705	0.6856	0.7762
	ERQ	1.1166	0.6510	1.8507	0.8603	0.6872	0.7779
	GPTQ	1.1084	0.6017	1.9216	0.7564	0.6849	0.7771
	SmoothQuant	1.0757	0.5948	2.0171	0.8664	0.6937	0.7853
	QuantVGGT	1.2870	1.1282	1.4357	0.8295	0.5286	0.5013
	TAPTQ	1.2694	0.7349	2.0161	1.0425	0.6775	0.7635
W8A8	FP	1.1855	0.7145	2.2195	1.3003	0.6893	0.7733
	RTN	1.1386	0.6696	2.2208	1.1332	0.6885	0.7745
	PTQ4ViT	1.1634	0.6921	2.0825	1.1920	0.6882	0.7726
	RepQ-ViT	1.0263	0.5870	1.7704	0.8771	0.6902	0.7792
	ERQ	1.0668	0.6092	1.9684	1.0211	0.6887	0.7757
	GPTQ	1.1317	0.6596	2.1534	1.1241	0.6888	0.7751
	SmoothQuant	1.1214	0.6456	2.0736	1.1374	0.6899	0.7764
	QuantVGGT	1.2829	1.1240	1.4370	0.8564	0.5295	0.5004
	TAPTQ	1.1649	0.6913	2.0890	1.1890	0.6917	0.7757

TABLE VIII

ETH3D COMPARISON UNDER THE REPAIRED 13-SEQUENCE $k_f = 5$ PROTOCOL. BASELINES USE THEIR STANDARD IMPLEMENTATIONS. FOR TAPTQ, EACH BIT-WIDTH REPORTS A REFERENCE AND A SELECTED HYPERPARAMETER CONFIGURATION; THE SELECTED ROW IS USED IN THE MAIN TABLE. LOWER ACC./COMP. AND HIGHER N.C. ARE BETTER. W6A6/W8A8 SELECTED CONFIGURATIONS USE FRESHLY CALIBRATED PROTOCOL-MATCHED CHECKPOINTS; THE INVALID HISTORICAL W6A6 RECORD IS EXCLUDED.

Bit	Method / TAPTQ configuration	Acc.↓		Comp.↓		N.C.↑	
		Mean	Med.	Mean	Med.	Mean	Med.
W4A8	FP	0.2909	0.1985	0.3628	0.2266	0.8433	0.9359
	RTN	0.5522	0.4284	0.6585	0.4059	0.7456	0.8581
	PTQ4ViT	0.5813	0.4490	0.7346	0.4119	0.7290	0.8403
	RepQ-ViT	0.8464	0.7299	2.0768	1.5578	0.5511	0.5829
	ERQ	0.5867	0.4540	0.7044	0.3986	0.7388	0.8470
	GPTQ	0.4322	0.3052	0.5030	0.3082	0.7779	0.8947
	SmoothQuant	0.6618	0.5428	0.8113	0.5057	0.7167	0.8182
	QuantVGGT	0.2765	0.1978	0.3557	0.2339	0.8405	0.9364
	TAPTQ reference ($\rho = .1, \tau = .007, r = 256$)	0.3608	0.2499	0.3947	0.2059	0.8079	0.9256
	TAPTQ selected ($\rho = .01, \tau = .005, r = 128$)	0.2816	0.1929	0.3457	0.1921	0.8371	0.9367
W6A6	FP	0.2909	0.1985	0.3628	0.2266	0.8433	0.9359
	RTN	0.5261	0.4035	0.6172	0.3913	0.7581	0.8597
	PTQ4ViT	0.4036	0.2869	0.4991	0.3136	0.7886	0.8951
	RepQ-ViT	0.5841	0.4404	0.6938	0.3752	0.7406	0.8457
	ERQ	0.5614	0.4335	0.6615	0.3596	0.7558	0.8631
	GPTQ	0.4758	0.3594	0.6833	0.4487	0.7518	0.8518
	SmoothQuant	0.6009	0.4515	0.7800	0.4911	0.7280	0.8258
	QuantVGGT	0.2929	0.2035	0.3674	0.2321	0.8434	0.9363
	TAPTQ reference ($\rho = .1, \tau = .007, r = 16$)	0.3870	0.2858	0.4463	0.2750	0.8003	0.9035
	TAPTQ selected ($\rho = .01, \tau = .005, r = 256$)	0.2811	0.1814	0.3287	0.1955	0.8434	0.9441
W8A8	FP	0.2909	0.1985	0.3628	0.2266	0.8433	0.9359
	RTN	0.3572	0.2593	0.5105	0.3423	0.7844	0.8819
	PTQ4ViT	0.2759	0.1779	0.3111	0.1863	0.8456	0.9470
	RepQ-ViT	0.6149	0.4299	0.5957	0.3401	0.7637	0.8844
	ERQ	0.5705	0.4403	0.6088	0.3654	0.7564	0.8681
	GPTQ	0.3718	0.2787	0.5056	0.3292	0.7836	0.8808
	SmoothQuant	0.3525	0.2628	0.4647	0.3018	0.7997	0.8879
	QuantVGGT	0.2908	0.1990	0.3650	0.2275	0.8422	0.9345
	TAPTQ reference ($\rho = .1, \tau = .007, r = 16$)	0.2786	0.1788	0.3127	0.1875	0.8443	0.9452
	TAPTQ selected ($\rho = .01, \tau = .005, r = 256$)	0.2743	0.1691	0.3146	0.1795	0.8455	0.9468

TABLE IX
ABLATION ON TAIL RATIO ρ UNDER W4A8. RESULTS ON 7SCENES.
FIXED $\tau = 0.007$ AND $r = 128$.

ρ	Acc.↓		Comp.↓		N.C.↑	
	Mean	Med.	Mean	Med.	Mean	Med.
0.001	0.027630	0.011376	0.045169	0.015677	0.688565	0.789571
0.005	0.023311	0.009792	0.040084	0.014786	0.691024	0.793678
0.010	0.021955	0.009290	0.038935	0.015268	0.691973	0.794978
0.020	0.021063	0.008780	0.037502	0.014833	0.693023	0.796225
0.050	0.019664	0.008336	0.035320	0.014718	0.693997	0.797692
0.100	0.018886	0.008108	0.035867	0.014019	0.694326	0.798332
0.500	0.020496	0.008527	0.037225	0.014545	0.692837	0.796400
1.000	0.023659	0.009530	0.039429	0.014456	0.689799	0.792253

TABLE X
ABLATION ON COMPENSATION THRESHOLD τ UNDER W4A8. RESULTS
ON 7SCENES. FIXED $\rho = 0.1$ AND $r = 128$.

τ	Acc.↓		Comp.↓		N.C.↑	
	Mean	Med.	Mean	Med.	Mean	Med.
0.001	0.020537	0.008571	0.037564	0.013989	0.692441	0.795647
0.003	0.022116	0.009014	0.037901	0.014260	0.692979	0.796600
0.005	0.020684	0.008571	0.037720	0.014213	0.693967	0.797300
0.007	0.018886	0.008108	0.035867	0.014019	0.694326	0.798332
0.010	0.020896	0.008649	0.037935	0.014396	0.692245	0.795451
0.020	0.028973	0.012740	0.048458	0.017451	0.685754	0.785780
0.050	0.028466	0.012369	0.049936	0.017173	0.690080	0.790282

TABLE XI
ABLATION ON LOW-RANK DIMENSION r UNDER W4A8. RESULTS ON
7SCENES. FIXED $\rho = 0.1$ AND $\tau = 0.007$.

Rank r	Acc.↓		Comp.↓		N.C.↑	
	Mean	Med.	Mean	Med.	Mean	Med.
8	0.027066	0.011533	0.045646	0.016145	0.690162	0.791503
16	0.025325	0.010828	0.045779	0.016523	0.691264	0.792966
32	0.020863	0.008915	0.039108	0.015731	0.692756	0.795568
64	0.019590	0.008517	0.037044	0.014933	0.692338	0.795237
128	0.018886	0.008108	0.035867	0.014019	0.694326	0.798332
256	0.018472	0.008117	0.033707	0.014021	0.694432	0.798279